

Pizza Sales Analysis

1. Project Overview

This project analyzes pizza sales data to understand ordering patterns, sales performance, product performance, and seasonality.

The analysis was guided by four main business questions:

How many customers do we have each day, and are there peak ordering hours?

How many pizzas are typically included in an order, and which pizzas are bestsellers?

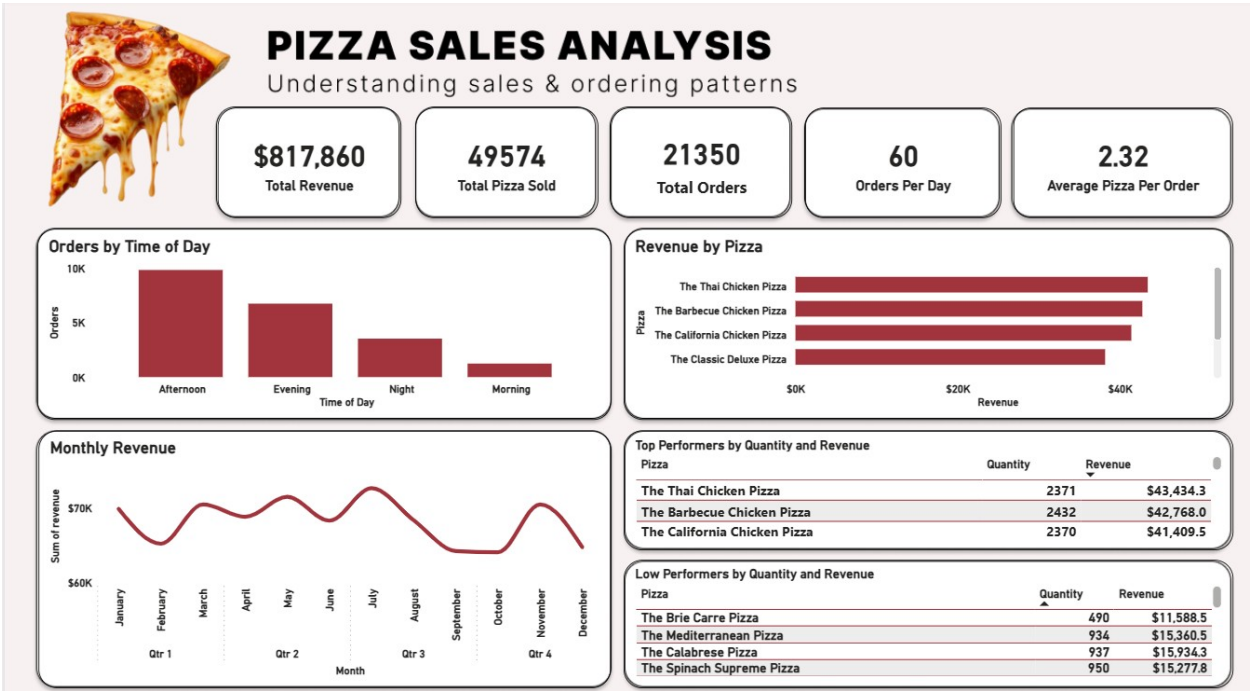
How much revenue was generated, and is there any seasonal pattern?

Are there pizzas that may need menu review, and are there opportunities for promotions?

Dashboard

Figure 1: Pizza Sales Analysis Dashboard

An interactive Power BI dashboard developed to analyze sales performance, ordering patterns, product performance, and seasonality.



2. Key Performance Indicators

Metric	Result
Total Revenue	\$817, 860
Total Pizzas Sold	49,574
Total Orders	21,350

Average Orders Per Day	60
Average Pizzas Per Order	2.3
Active Ordering Days	358

Note: The dataset does not contain a customer identifier, so "customers per day" cannot be directly calculated. The analysis therefore uses orders as the measurable activity.

3. Ordering Patterns

Are there peak ordering hours?

Yes. Orders vary considerably by time of day.

Afternoon: ~9.8K orders

Evening: ~6.7K

Night: ~3.5K

Morning: ~1.2K

Afternoon recorded the highest number of orders, while morning had the lowest.

This suggests that customer ordering activity is concentrated later in the day, particularly during the afternoon.

4. Pizzas per Order & Bestsellers

Customers purchased an average of 2.32 pizzas per order.

The best-performing pizzas were identified by examining both quantity sold and revenue generated.

The analysis showed that several of the highest-selling pizzas also generated substantial revenue, including:

The Barbecue Chicken Pizza

The California Chicken Pizza

The Classic Deluxe Pizza

The Thai Chicken Pizza

Looking at both quantity and revenue provides a better picture of product performance than relying on either metric alone.

5. Revenue & Seasonality

The business generated approximately \$817,860 in total revenue.

Monthly revenue showed noticeable fluctuations throughout the year.

July recorded the strongest monthly revenue, while revenue declined considerably after the peak, reaching its lowest levels around September/October before recovering in November.

This indicates a noticeable seasonal pattern in the dataset.

6. Menu Performance

To identify pizzas that may require further review, I compared each pizza's total quantity sold and total revenue against the median performance across pizzas.

The median benchmarks were:

Median quantity: 1,451.5 pizzas

Median revenue: \$24,734.38

Pizzas falling below both benchmarks were treated as low performers.

Examples include:

Pizza	Quantity	Revenue
The Brie Carré Pizza	490	\$11,588.50
The Mediterranean Pizza	934	\$15,360.50
The Calabrese Pizza	937	\$15,934.25

These results don't automatically mean these pizzas should be removed from the menu. They identify products that may warrant further investigation.

7. Potential Business Opportunities

The analysis suggests two areas worth exploring:

Menu Review

Low-performing pizzas could be investigated further to understand whether their performance is related to pricing, customer preferences, or other factors before making menu decisions.

Promotional Opportunities

Since July was the strongest sales period and revenue declined afterward, targeted promotions during weaker periods could be explored to help sustain demand.

These are data-informed opportunities rather than conclusions about causation; additional information such as promotional history, costs, and customer-level data would be needed to evaluate them properly.

8. Tools Used

Power BI — Dashboard development and visualization

DAX — Measures and calculations

Power Query — Data preparation

Figma — Dashboard UI/design concept

Final Takeaway

The analysis revealed that sales performance is influenced by when customers order, which pizzas they purchase, and how demand changes throughout the year.